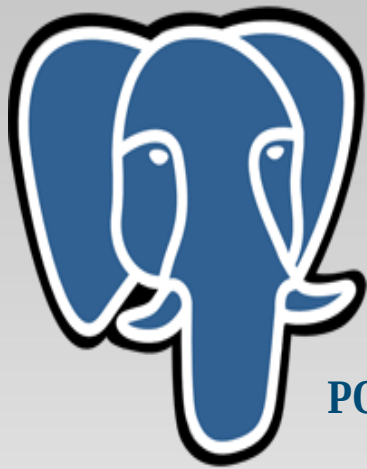




PostgreSQL

POSTGRES DATABASE ADMINISTRATION



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[SENIOR DATABASE ADMINISTRATOR|ORACLE| POSTGRESQL | MSSQL|OCI|AWS]

CHAPTER 3: POSTGRESQL PHYSICAL STRUCTURES – DIRECTORY STRUCTURES.

About the Author:

I am **Stephen Njorge**, a Senior Database Engineer with over 8+ years of expertise in designing, securing and optimizing high-performance database systems across the banking and fintech sectors. My core technical proficiency spans major databases: Oracle, PostgreSQL, MS SQL Server and MySQL-along with advanced scripting in SQL and Python for automation and data pipeline development. I architect and manage cloud and virtualization solutions on OCI, AWS, and Proxmox for onprem systems and implement comprehensive DevOps practices using monitoring tools like Prometheus and Grafana and IT service management via iTop. Further, I deliver advanced data analytics and visualization using Tableau, Power BI and Oracle BI. As a Certified Data Protection Officer, I integrate robust data protection, compliance and security measures including vulnerability assessment and role-based access control into every layer of the data infrastructure.

Chapter 3: PostgreSQL Physical Structures - Directory Structures.

Understanding PostgreSQL service architecture:

PostgreSQL uses a process-based, client/server architecture designed for reliability, concurrency and extensibility. Below is a clear, DBA-oriented explanation from the moment a client connects to how data is stored and managed.

Overview

PostgreSQL consists of:

- **Client applications** (psql, pgAdmin, applications)
- **PostgreSQL server (postmaster / postgres)**
- **Backend server processes**
- **Shared memory**
- **Background processes**
- **Physical storage (data files, WAL, configs)**

Client–Server Model

- PostgreSQL runs as a **server process** (`postmaster` or `postgres`)
- Clients connect over:
- TCP/IP
- Unix domain sockets
- Each client connection is handled by **a dedicated backend process**

Postmaster (Parent Process)

The **postmaster** process:

- Starts when PostgreSQL is launched
- Listens for incoming connections
- Authenticates users
- Forks a new backend process for each connection
- Manages shared memory and background workers

Backend Server Processes

Each client gets its own backend process, which:

- Parses SQL
- Generates query plans
- Executes queries
- Reads/writes data via shared buffers
- Communicates with background processes

Shared Memory Architecture

PostgreSQL uses **shared memory** for coordination and performance.

Key Shared Memory Areas:

a) Shared Buffers

- Cache of table and index pages
- Reduces disk I/O
- Controlled by shared_buffers

b) WAL Buffers

- Temporarily store WAL records
- Written to disk by WAL writer

c) Lock Manager

- Manages table, row and transaction locks

d) Process Information

- Active sessions
- Transaction status
- Statistics

Background Processes

PostgreSQL relies on several background processes to maintain performance and consistency.

Key Background Processes:

<u>Process</u>	<u>Role</u>
Checkpointer	Writes dirty buffers to disk at checkpoints
Background Writer	Gradually writes dirty pages
WAL Writer	Flushes WAL buffers to disk
Autovacuum Launcher	Starts autovacuum workers
Autovacuum Worker	Cleans dead tuples
Stats Collector	Collects performance stats
Logical Replication Workers	Handle logical replication

Query Processing Flow

When a client runs a query:

1. **Parser** – checks syntax
2. **Analyzer** – resolves tables, columns, types
3. **Planner/Optimizer** – chooses the best execution plan
4. **Executor** – runs the plan
5. **Results returned to client**

Transaction and MVCC Architecture

PostgreSQL uses **MVCC (Multi-Version Concurrency Control)**:

- Readers don't block writers
- Writers don't block readers
- Each row version has:
 - xmin (creating transaction)
 - xmax (deleting/updating transaction)

Write-Ahead Logging (WAL)

To guarantee durability:

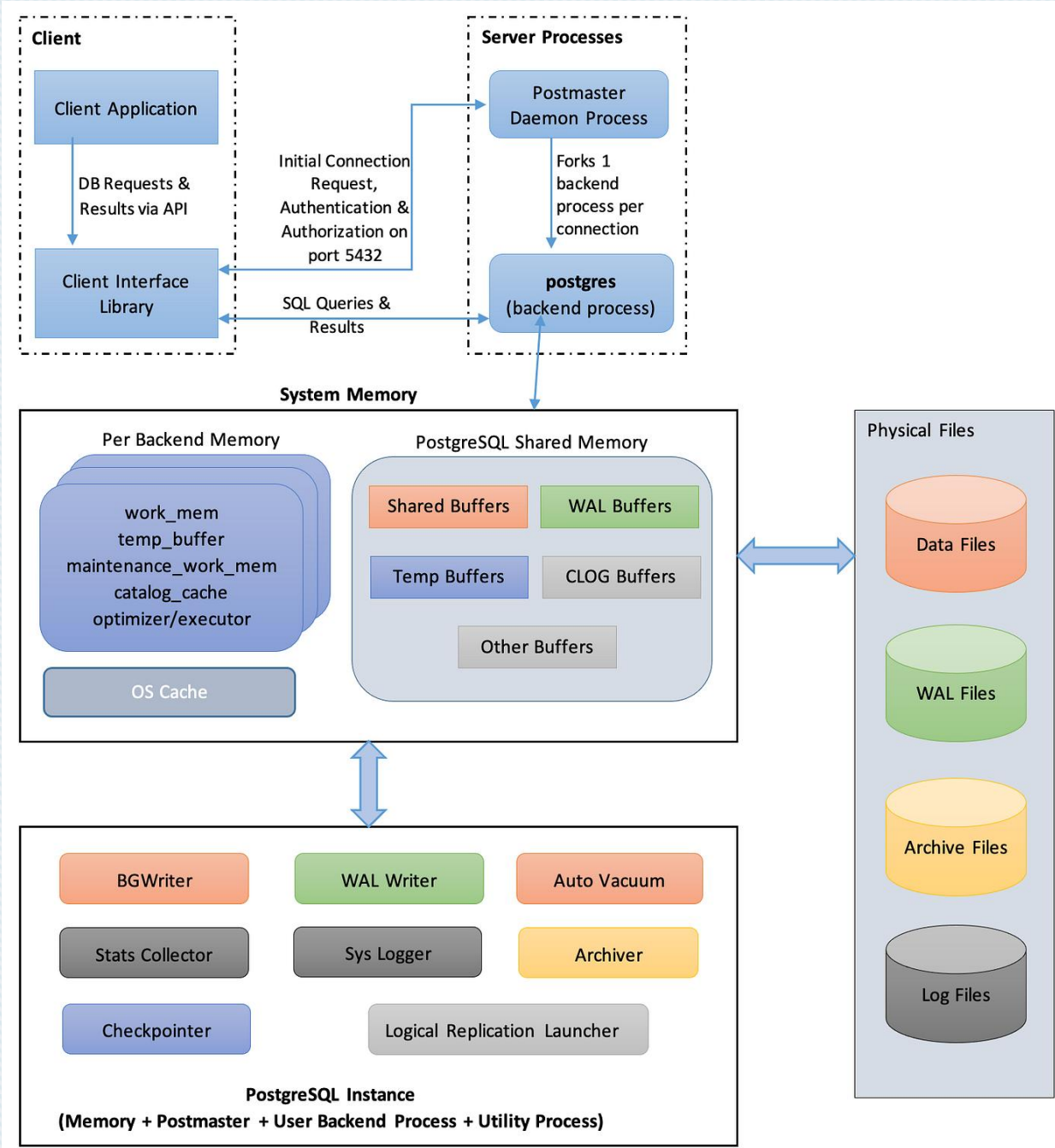
- Changes are first written to **WAL**
- Data files are updated later
- WAL files live in `pg_wal/`

Physical Storage Layout

Inside the data directory:

<code>base/</code>	→ database data files
<code>pg_wal/</code>	→ Write-Ahead Logs
<code>pg_tblspc/</code>	→ Tablespaces
<code>global/</code>	→ cluster-wide tables
<code>pg_stat/</code>	→ statistics

PostgreSQL Server Architecture.



Directory Layout

```
/ (Root)
├── /var/lib/postgresql/
│   ├── /18/
│   │   ├── main/
│   │   │   ├── base/
│   │   │   │   ├── 1/
│   │   │   │   ├── 13427/
│   │   │   │   └── 13428/
│   │   │   ├── global/
│   │   │   ├── pg_wal/
│   │   │   ├── pg_multixact/
│   │   │   ├── pg_commit_ts/
│   │   │   ├── pg_dynshmem/
│   │   │   ├── pg_logical/
│   │   │   ├── pg_notify/
│   │   │   ├── pg_replslot/
│   │   │   ├── pg_serial/
│   │   │   ├── pg_snapshots/
│   │   │   ├── pg_stat/
│   │   │   ├── pg_stat_tmp/
│   │   │   ├── pg_subtrans/
│   │   │   ├── pg_tblspc/
│   │   │   ├── pg_twophase/
│   │   │   └── PG_VERSION
│   │   └── othercluster/
│   ├── /13/
│   └── /15/
├── /etc/postgresql/
│   ├── /18/
│   │   ├── main/
│   │   │   ├── postgresql.conf
│   │   │   ├── pg_hba.conf
│   │   │   └── pg_ident.conf
│   │   └── othercluster/
│   └── /15/main/
├── /var/log/postgresql/
│   ├── postgresql-18-main.log
│   └── postgresql-15-main.log
├── /tmp/
├── /home/postgres/
└── /mnt/postgres_data/

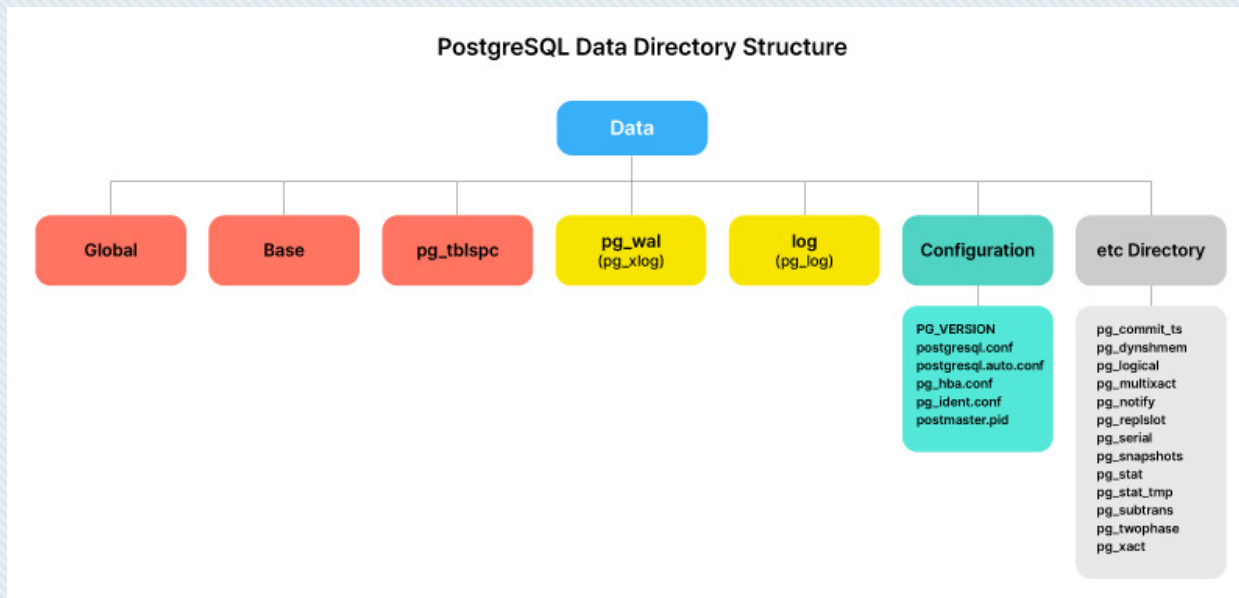
# Main data directory (default)
# Version-specific directory
# Default cluster "main"
# Database files (critical)
# System database (template1)
# User database 1
# User database 2
# Cluster-wide tables
# Write-Ahead Logs (WAL) [previously pg_xlog]
# MultiXact data
# Commit timestamps
# Dynamic shared memory
# Logical decoding
# LISTEN/NOTIFY data
# Replication slots
# Serializable transaction info
# Exported snapshots
# Permanent statistics
# Temporary statistics
# Subtransaction data
# Tablespace symlinks
# Prepared transactions
# Version file
# Additional cluster
# Another PostgreSQL version
# Latest version
# Configuration files
# Main config
# Client auth
# User mapping
# Log files
# Temporary files
# Postgres user home
# Alternative data mount (for large DBs)
```

Enterprise-Grade Production Layout

```
/ (Root)
├── /postgres/
│   ├── /data/
│   │   ├── /18/main/
│   │   ├── /18/replica/
│   │   └── /backups/
│   ├── /wal/
│   ├── /tablespaces/
│   │   ├── /indexes/
│   │   ├── /hot_data/
│   │   └── /cold_data/
│   ├── /archive/
│   ├── /backups/
│   └── /scripts/
├── /etc/postgresql/
└── /var/log/postgresql/
```

Dedicated PostgreSQL mount point
Main data (fast SSD/NVMe)
Current production
Replica data
Base backups
WAL directory (separate fast disk)
Tablespaces (optional)
Index tablespace
Frequently accessed data
Archived/rarely accessed
Archived WALs (large HDD)
Logical backups
Maintenance scripts
Configuration
Logs

Data Directory Structure:



How to Verify Directory Structure

Check Current Data Directory

-- From inside PostgreSQL

SHOW data_directory;

```
postgres=# SHOW data_directory;
          data_directory
```

```
-----
 /var/lib/postgresql/18/dev
(1 row)
```

```
postgres=#
```

SHOW config_file;

```
postgres=# SHOW config_file;
          config_file
```

```
-----
 /etc/postgresql/18/dev/postgresql.conf
(1 row)
```

```
postgres=#
```

SHOW hba_file;

```
postgres=# SHOW hba_file;
          hba_file
```

```
-----
 /etc/postgresql/18/dev/pg_hba.conf
(1 row)
```

```
postgres=# |
```

-- Get all directories

```
SELECT name, setting FROM pg_settings
WHERE name LIKE '%dir%' OR name LIKE '%file%'
ORDER BY name;
```

```
postgres=# SELECT name, setting FROM pg_settings
WHERE name LIKE '%dir%' OR name LIKE '%file%'
ORDER BY name;
```

name	setting
config_file	/etc/postgresql/18/dev/postgresql.conf
data_directory	/var/lib/postgresql/18/dev
data_directory_mode	0700
debug_io_direct	
external_pid_file	/var/run/postgresql/18-dev.pid
file_copy_method	copy
hba_file	/etc/postgresql/18/dev/pg_hba.conf
ident_file	/etc/postgresql/18/dev/pg_ident.conf
krb_server_keyfile	FILE:/etc/postgresql-common/krb5.keytab
log_directory	log
log_file_mode	0600
log_filename	postgresql-%Y-%m-%d_%H%M%S.log
log_temp_files	-1
max_files_per_process	1000
remove_temp_files_after_crash	on
ssl_ca_file	
ssl_cert_file	/etc/ssl/certs/ssl-cert-snakeoil.pem
ssl_crl_dir	
ssl_crl_file	
ssl_dh_params_file	
ssl_key_file	/etc/ssl/private/ssl-cert-snakeoil.key
temp_file_limit	-1
unix_socket_directories	/var/run/postgresql
vacuum_cost_page_dirty	20

(24 rows)

```
postgres=#
```

Verify File system Structure

```
# Check main data directory
```

```
ls -la /var/lib/postgresql/18/main
```

```
postgres@998-986-02:/home/steve$ ls -la /var/lib/postgresql/18/main
total 96
drwx----- 20 postgres postgres 4096 Jan 14 10:41 .
drwxr-xr-x  6 postgres postgres 4096 Jan 13 22:05 ..
-rw-----  1 postgres postgres   3 Dec 21 19:13 PG_VERSION
drwx-----  9 postgres postgres 4096 Dec 23 16:46 base
-rw-----  1 postgres postgres  48 Jan 14 10:41 current_logfiles
drwx-----  2 postgres postgres 4096 Jan 13 21:43 global
drwx-----  2 postgres postgres 4096 Jan 14 10:41 log
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_commit_ts
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_dynshmem
drwx-----  4 postgres postgres 4096 Jan 13 21:43 pg_logical
drwx-----  4 postgres postgres 4096 Dec 21 19:13 pg_multixact
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_notify
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_replslot
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_serial
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_snapshots
drwx-----  2 postgres postgres 4096 Dec 21 21:16 pg_stat
drwx-----  2 postgres postgres 4096 Jan 14 10:41 pg_stat_tmp
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_subtrans
drwx-----  2 postgres postgres 4096 Dec 23 17:27 pg_tblspc
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_twophase
drwx-----  4 postgres postgres 4096 Dec 23 16:46 pg_wal
drwx-----  2 postgres postgres 4096 Dec 21 19:13 pg_xact
-rw-----  1 postgres postgres 117 Dec 23 17:28 postgresql.auto.conf
-rw-----  1 postgres postgres 130 Jan 13 21:43 postmaster.opts
postgres@998-986-02:/home/steve$
```

```
# Check permissions
```

```
ls -la /var/lib/postgresql/18/main/ | head -20
```

```
postgres@998-986-02:/home/steve$ ls -la /var/lib/postgresql/18/main/ | head -20
total 96
drwx----- 20 postgres postgres 4096 Jan 14 10:41 .
drwxr-xr-x   6 postgres postgres 4096 Jan 13 22:05 ..
-rw-----   1 postgres postgres   3 Dec 21 19:13 PG_VERSION
drwx-----   9 postgres postgres 4096 Dec 23 16:46 base
-rw-----   1 postgres postgres  48 Jan 14 10:41 current_logfiles
drwx-----   2 postgres postgres 4096 Jan 13 21:43 global
drwx-----   2 postgres postgres 4096 Jan 14 10:41 log
drwx-----   2 postgres postgres 4096 Dec 21 19:13 pg_commit_ts
drwx-----   2 postgres postgres 4096 Dec 21 19:13 pg_dynshmem
drwx-----   4 postgres postgres 4096 Jan 13 21:43 pg_logical
drwx-----   4 postgres postgres 4096 Dec 21 19:13 pg_multixact
drwx-----   2 postgres postgres 4096 Dec 21 19:13 pg_notify
drwx-----   2 postgres postgres 4096 Dec 21 19:13 pg_replslot
drwx-----   2 postgres postgres 4096 Dec 21 19:13 pg_serial
drwx-----   2 postgres postgres 4096 Dec 21 19:13 pg_snapshots
drwx-----   2 postgres postgres 4096 Dec 21 21:16 pg_stat
drwx-----   2 postgres postgres 4096 Jan 14 10:41 pg_stat_tmp
drwx-----   2 postgres postgres 4096 Dec 21 19:13 pg_subtrans
drwx-----   2 postgres postgres 4096 Dec 23 17:27 pg_tblspc
postgres@998-986-02:/home/steve$
```

Verify whether the Key Directories Exists or not.

```
#!/bin/bash
```

```
# verify_structure.sh
```

```
DATA_DIR=$(psql -t -c "SHOW data_directory;" | xargs)
```

```
echo "Data Directory: $DATA_DIR"
```

```
echo "Checking essential directories..."
```

```
for dir in base global pg_wal pg_xact pg_multixact pg_commit_ts; do
```

```
    if [ -d "$DATA_DIR/$dir" ]; then
```

```
        echo "✓ $dir exists"
```

```
    else
```

```
        echo "✗ $dir MISSING!"
```

```
    fi
```

```
done
```

```
echo -e "\nChecking permissions..."
```

```
ls -ld $DATA_DIR
```

```
ls -ld $DATA_DIR/base
```

```

postgres@998-986-02:/home/steve$ #!/bin/bash
# verify_structure.sh

DATA_DIR=$(psql -t -c "SHOW data_directory;" | xargs)
echo "Data Directory: $DATA_DIR"

echo "Checking essential directories..."
for dir in base global pg_wal pg_xact pg_multixact pg_commit_ts; do
    if [ -d "$DATA_DIR/$dir" ]; then
        echo "✓ $dir exists"
    else
        echo "✗ $dir MISSING!"
    fi
done

echo -e "\nChecking permissions..."
ls -ld $DATA_DIR
ls -ld $DATA_DIR/base
Data Directory: /var/lib/postgresql/18/dev
Checking essential directories...
✓ base exists
✓ global exists
✓ pg_wal exists
✓ pg_xact exists
✓ pg_multixact exists
✓ pg_commit_ts exists

Checking permissions...
drwx----- 19 postgres postgres 4096 Jan 14 10:41 /var/lib/postgresql/18/dev
drwx----- 5 postgres postgres 4096 Jan 13 21:52 /var/lib/postgresql/18/dev/base
postgres@998-986-02:/home/steve$

```

Check Directory Permissions

```

# ALL should be owned by postgres user
ls -la /var/lib/postgresql/18/main/
# Files: 600 or 640, Directories: 700 or 750

```

```

postgres@998-986-02:/home/steve$ ls -la /var/lib/postgresql/18/main/
total 96
drwx----- 20 postgres postgres 4096 Jan 14 10:41 .
drwxr-xr-x 6 postgres postgres 4096 Jan 13 22:05 ..
-rw----- 1 postgres postgres 3 Dec 21 19:13 PG_VERSION
drwx----- 9 postgres postgres 4096 Dec 23 16:46 base
-rw----- 1 postgres postgres 48 Jan 14 10:41 current_logfiles
drwx----- 2 postgres postgres 4096 Jan 13 21:43 global
drwx----- 2 postgres postgres 4096 Jan 14 10:41 log
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_commit_ts
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_dynshmem
drwx----- 4 postgres postgres 4096 Jan 13 21:43 pg_logical
drwx----- 4 postgres postgres 4096 Dec 21 19:13 pg_multixact
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_notify
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_replslot
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_serial
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_snapshots
drwx----- 2 postgres postgres 4096 Dec 21 21:16 pg_stat
drwx----- 2 postgres postgres 4096 Jan 14 10:41 pg_stat_tmp
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_subtrans
drwx----- 2 postgres postgres 4096 Dec 23 17:27 pg_tblspc
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_twophase
drwx----- 4 postgres postgres 4096 Dec 23 16:46 pg_wal
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_xact
-rw----- 1 postgres postgres 117 Dec 23 17:28 postgresql.auto.conf
-rw----- 1 postgres postgres 130 Jan 13 21:43 postmaster.opts
postgres@998-986-02:/home/steve$

```

Verify Disk Layout

```
# Check mount points
df -h /var/lib/postgresql
```

```
postgres@998-986-02:/home/steve$ df -h /var/lib/postgresql
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdc         1007G  2.7G  953G   1% /
postgres@998-986-02:/home/steve$
```

```
df -h /var/log/postgresql
```

```
postgres@998-986-02:/home/steve$ df -h /var/log/postgresql
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdc         1007G  2.7G  953G   1% /
postgres@998-986-02:/home/steve$
```

```
# Check disk usage
du -sh /var/lib/postgresql/18/main/*
```

```
postgres@998-986-02:/home/steve$ du -sh /var/lib/postgresql/18/main/*
4.0K    /var/lib/postgresql/18/main/PG_VERSION
54M     /var/lib/postgresql/18/main/base
4.0K    /var/lib/postgresql/18/main/current_logfiles
556K    /var/lib/postgresql/18/main/global
1.7M    /var/lib/postgresql/18/main/log
4.0K    /var/lib/postgresql/18/main/pg_commit_ts
4.0K    /var/lib/postgresql/18/main/pg_dynshmem
16K     /var/lib/postgresql/18/main/pg_logical
28K     /var/lib/postgresql/18/main/pg_multixact
4.0K    /var/lib/postgresql/18/main/pg_notify
4.0K    /var/lib/postgresql/18/main/pg_replslot
4.0K    /var/lib/postgresql/18/main/pg_serial
4.0K    /var/lib/postgresql/18/main/pg_snapshots
4.0K    /var/lib/postgresql/18/main/pg_stat
4.0K    /var/lib/postgresql/18/main/pg_stat_tmp
12K     /var/lib/postgresql/18/main/pg_subtrans
4.0K    /var/lib/postgresql/18/main/pg_tblspc
4.0K    /var/lib/postgresql/18/main/pg_twophase
65M     /var/lib/postgresql/18/main/pg_wal
12K     /var/lib/postgresql/18/main/pg_xact
4.0K    /var/lib/postgresql/18/main/postgresql.auto.conf
4.0K    /var/lib/postgresql/18/main/postmaster.opts
postgres@998-986-02:/home/steve$ |
```

```
# Check inode usage
df -i /var/lib/postgresql
```

```
postgres@998-986-02:/home/steve$ df -i /var/lib/postgresql
Filesystem      Inodes IUsed   IFree IUse% Mounted on
/dev/sdc        67108864 67308 67041556    1% /
postgres@998-986-02:/home/steve$ |
```

Configuration Validation

```
# Check if config files exist
ls -la /etc/postgresql/18/main/
# Should have: postgresql.conf, pg_hba.conf, pg_ident.conf
```

```
postgres@998-986-02:/home/steve$ ls -la /etc/postgresql/18/main/
total 108
drwxr-xr-x 3 postgres postgres 4096 Dec 21 21:15 .
drwxr-xr-x 5 postgres postgres 4096 Jan 13 22:05 ..
drwxr-xr-x 2 postgres postgres 4096 Dec 21 19:13 conf.d
-rw-r--r-- 1 postgres postgres 315 Dec 21 19:13 environment
-rw-r--r-- 1 postgres postgres 143 Dec 21 19:13 pg_ctl.conf
-rw-r----- 1 postgres postgres 5924 Dec 21 19:13 pg_hba.conf
-rw-r----- 1 root      root      5924 Dec 21 20:54 pg_hba.conf.backup
-rw-r----- 1 postgres postgres 2640 Dec 21 19:13 pg_ident.conf
-rw-r--r-- 1 postgres postgres 30374 Dec 21 19:13 postgresql.conf
-rw-r--r-- 1 root      root      30374 Dec 21 20:54 postgresql.conf.backup
-rw-r--r-- 1 postgres postgres 317 Dec 21 19:13 start.conf
postgres@998-986-02:/home/steve$
```

```
# Verify config syntax
sudo -u postgres /usr/lib/postgresql/18/bin/postgres \
  --config-file=/etc/postgresql/18/main/postgresql.conf \
  --check
```

Complete Health Check Script

```
#!/bin/bash
# postgres_health_check.sh

echo "=== PostgreSQL Health Check ==="
echo "Timestamp: $(date)"
echo ""

# 1. System Information
echo "1. System Info:"
echo "Hostname: $(hostname)"
echo "PostgreSQL Version: $(psql --version)"
echo ""

# 2. Running Clusters
echo "2. Running Clusters:"
```

```

if command -v pg_lsclusters &> /dev/null; then
    sudo pg_lsclusters
else
    ps aux | grep postgres | grep -v grep
fi
echo ""

# 3. Data Directory Structure
echo "3. Data Directory Structure:"
PGDATA=$(psql -t -c "SHOW data_directory;" 2>/dev/null | xargs)
if [ -z "$PGDATA" ]; then
    PGDATA="/var/lib/postgresql"
fi

echo "PGDATA: $PGDATA"
if [ -d "$PGDATA" ]; then
    tree -L 3 $PGDATA 2>/dev/null || find $PGDATA -maxdepth 3 -type d | sort
else
    echo "ERROR: PGDATA directory not found!"
fi
echo ""

# 4. Permissions Check
echo "4. Permission Check:"
sudo find $PGDATA -type f ! -user postgres 2>/dev/null | head -5
if [ $? -eq 0 ]; then
    echo "WARNING: Files not owned by postgres user!"
else
    echo "✓ All files owned by postgres"
fi
echo ""

# 5. Disk Usage
echo "5. Disk Usage:"
df -h $PGDATA
echo ""
du -sh $PGDATA/* 2>/dev/null | sort -hr
echo ""

# 6. Configuration Files
echo "6. Configuration Files:"
CONF_FILE=$(psql -t -c "SHOW config_file;" 2>/dev/null | xargs)
HBA_FILE=$(psql -t -c "SHOW hba_file;" 2>/dev/null | xargs)

echo "Config: $CONF_FILE"
echo "HBA: $HBA_FILE"
if [ -f "$CONF_FILE" ]; then
    echo "Config size: $(wc -l < "$CONF_FILE") lines"
fi
echo ""

# 7. Database Status
echo "7. Database Status:"
psql -c "SELECT datname, pg_size_pretty(pg_database_size(datname)) as size,
        datcollate, datctype FROM pg_database ORDER BY size DESC;"

```

```
postgres@998-986-02:/home/steve$ ./postgres_health_check.sh
=== PostgreSQL Health Check ===
Timestamp: Wed Jan 14 15:00:14 MSK 2026

1. System Info:
Hostname: 998-986-02
PostgreSQL Version: psql (PostgreSQL) 18.1 (Ubuntu 18.1-1.pgdg24.04+2)

2. Running Clusters:
[sudo] password for postgres:
postgres is not in the sudoers file.

3. Data Directory Structure:
PGDATA: /var/lib/postgresql/18/dev
/var/lib/postgresql/18/dev
/var/lib/postgresql/18/dev/base
/var/lib/postgresql/18/dev/base/1
/var/lib/postgresql/18/dev/base/4
/var/lib/postgresql/18/dev/base/5
/var/lib/postgresql/18/dev/global
/var/lib/postgresql/18/dev/pg_commit_ts
/var/lib/postgresql/18/dev/pg_dynshmem
/var/lib/postgresql/18/dev/pg_logical
/var/lib/postgresql/18/dev/pg_logical/mappings
/var/lib/postgresql/18/dev/pg_logical/snapshots
/var/lib/postgresql/18/dev/pg_multixact
/var/lib/postgresql/18/dev/pg_multixact/members
/var/lib/postgresql/18/dev/pg_multixact/offsets
/var/lib/postgresql/18/dev/pg_notify
/var/lib/postgresql/18/dev/pg_replslot
/var/lib/postgresql/18/dev/pg_serial
/var/lib/postgresql/18/dev/pg_snapshots
/var/lib/postgresql/18/dev/pg_stat
/var/lib/postgresql/18/dev/pg_stat_tmp
/var/lib/postgresql/18/dev/pg_subtrans
```

```
/var/lib/postgresql/18/dev/pg_subtrans
/var/lib/postgresql/18/dev/pg_tblspc
/var/lib/postgresql/18/dev/pg_twophase
/var/lib/postgresql/18/dev/pg_wal
/var/lib/postgresql/18/dev/pg_wal/archive_status
/var/lib/postgresql/18/dev/pg_wal/summaries
/var/lib/postgresql/18/dev/pg_xact
```

4. Permission Check:

```
[sudo] password for postgres:
```

```
WARNING: Files not owned by postgres user!
```

5. Disk Usage:

```
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdc        1007G  2.7G  953G   1% /
```

```
23M    /var/lib/postgresql/18/dev/base
17M    /var/lib/postgresql/18/dev/pg_wal
556K   /var/lib/postgresql/18/dev/global
28K    /var/lib/postgresql/18/dev/pg_multixact
16K    /var/lib/postgresql/18/dev/pg_logical
12K    /var/lib/postgresql/18/dev/pg_xact
12K    /var/lib/postgresql/18/dev/pg_subtrans
4.0K   /var/lib/postgresql/18/dev/postmaster.pid
4.0K   /var/lib/postgresql/18/dev/postmaster.opts
4.0K   /var/lib/postgresql/18/dev/postgresql.auto.conf
4.0K   /var/lib/postgresql/18/dev/pg_twophase
4.0K   /var/lib/postgresql/18/dev/pg_tblspc
4.0K   /var/lib/postgresql/18/dev/pg_stat_tmp
4.0K   /var/lib/postgresql/18/dev/pg_stat
4.0K   /var/lib/postgresql/18/dev/pg_snapshots
4.0K   /var/lib/postgresql/18/dev/pg_serial
4.0K   /var/lib/postgresql/18/dev/pg_replslot
4.0K   /var/lib/postgresql/18/dev/pg_notify
```

```
4.0K   /var/lib/postgresql/18/dev/pg_dynshmem
4.0K   /var/lib/postgresql/18/dev/pg_commit_ts
4.0K   /var/lib/postgresql/18/dev/PG_VERSION
```

6. Configuration Files:

```
Config: /etc/postgresql/18/dev/postgresql.conf
```

```
HBA: /etc/postgresql/18/dev/pg_hba.conf
```

```
Config size: 884 lines
```

7. Database Status:

datname	size	datcollate	datctype
template1	7742 kB	C.UTF-8	C.UTF-8
postgres	7670 kB	C.UTF-8	C.UTF-8
template0	7513 kB	C.UTF-8	C.UTF-8

(3 rows)

```
postgres@998-986-02: /home/steve$
```

Check Tablespace Structure

```
-- View tablespaces and their locations
SELECT spcname, pg_tablespace_location(oid)
FROM pg_tablespace;
```

```
postgres=# SELECT spcname, pg_tablespace_location(oid)
FROM pg_tablespace;
 spcname | pg_tablespace_location
-----|-----
 pg_default |
 pg_global |
 ts1      | /var/lib/pgsql/tablespaces/ts1
 ts2      | /var/lib/pgsql/tablespaces/ts2
 ts3      | /var/lib/pgsql/tablespaces/ts3
 ts4      | /var/lib/pgsql/tablespaces/ts4
 tspace1   | /u01/pgsql16/pgdata/tablespace1
 temp_ts   | /u01/pgsql16/pgdata/temp_files
(8 rows)

postgres=# |
```

```
-- Check symlinks in pg_tblspc
ls -la /var/lib/postgresql/18/main/pg_tblspc/
```

```
postgres@998-986-02:/home/steve$ ls -la /var/lib/postgresql/18/main/pg_tblspc/
total 8
drwx----- 2 postgres postgres 4096 Dec 23 17:27 .
drwx----- 20 postgres postgres 4096 Jan 14 15:36 ..
lrwxrwxrwx 1 postgres postgres 30 Dec 21 19:13 16385 -> /var/lib/pgsql/tablespaces/ts1
lrwxrwxrwx 1 postgres postgres 30 Dec 21 19:13 16386 -> /var/lib/pgsql/tablespaces/ts2
lrwxrwxrwx 1 postgres postgres 30 Dec 21 19:13 16387 -> /var/lib/pgsql/tablespaces/ts3
lrwxrwxrwx 1 postgres postgres 30 Dec 21 19:13 16388 -> /var/lib/pgsql/tablespaces/ts4
lrwxrwxrwx 1 postgres postgres 31 Dec 23 16:34 24848 -> /u01/pgsql16/pgdata/tablespace1
lrwxrwxrwx 1 postgres postgres 30 Dec 23 17:27 24855 -> /u01/pgsql16/pgdata/temp_files
postgres@998-986-02:/home/steve$
```

WAL Directory Verification

```
# Check WAL directory
ls -la /var/lib/postgresql/18/main/pg_wal/
```

```
postgres@998-986-02:/home/steve$ ls -la /var/lib/postgresql/18/main/pg_wal/
total 65552
drwx----- 4 postgres postgres 4096 Dec 23 16:46 .
drwx----- 20 postgres postgres 4096 Jan 14 15:36 ..
-rw----- 1 postgres postgres 16777216 Dec 21 19:13 00000001000000000000000001
-rw----- 1 postgres postgres 16777216 Dec 21 21:16 00000001000000000000000002
-rw----- 1 postgres postgres 16777216 Dec 23 16:48 00000001000000000000000003
-rw----- 1 postgres postgres 16777216 Jan 14 15:37 00000001000000000000000004
drwx----- 2 postgres postgres 4096 Dec 23 16:48 archive_status
drwx----- 2 postgres postgres 4096 Dec 21 19:13 summaries
postgres@998-986-02:/home/steve$
```

Should contain .wal files

Check WAL archive status

```
psql -c "SELECT * FROM pg_stat_archiver;"
```

archived_count	last_archived_wal	last_archived_time	failed_count	last_failed_wal	last_failed_time	stats_reset
0			9	00000001000000000000000001	2026-01-14 15:39:06.457277+03	2026-01-14 15:36:48.628198+03

(1 row)

PostgreSQL Configuration scripts for Ideal Structure

```
# postgresql.conf
data_directory = '/postgres/data/18/main'
hba_file = '/etc/postgresql/18/main/pg_hba.conf'
ident_file = '/etc/postgresql/18/main/pg_ident.conf'
external_pid_file = '/var/run/postgresql/18-main.pid'

# Logging
log_directory = '/var/log/postgresql'
log_filename = 'postgresql-18-main.log'

# WAL Settings
wal_level = replica
archive_mode = on
archive_command = 'cp %p /postgres/archive/%f'
wal_keep_size = 10GB
```